

Lavkush Gupta

Senior AI Engineer · LLM Agents · Computer Vision · Deep Learning

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SUMMARY

Senior AI Engineer with 4+ years building production LLM, computer-vision and ML/DL systems that replace manual operations. At Khetika, shipped three 0-to-1 products used by 100+ staff across 2000+ SKUs, cutting reporting time 80% and automating quality control on edge hardware. Specialized in RAG, agents and high-FPS computer vision on Jetson, with ownership across the full arc: annotation, training, deployment, APIs and the MLOps that keeps it running. **Patent granted · FICCI Award 2025.**

EXPERIENCE

Senior AI Engineer · Khetika (SuperZop)

Feb 2023 – Present · Mumbai

Promoted from Machine Learning Engineer; led three production AI products end-to-end.

SuperTRACE: LLM SUPPLY-CHAIN PLATFORM · LEAD · FICCI AWARD 2025

- Architected an NL-to-SQL agent with schema-grounding and query validation, eliminating 120+ hours/month of manual reporting.
- Adopted by 100+ users across warehouse, finance and ops for 2000+ SKUs; cut data-entry errors from 5–10/day to under 2/week.
- Deployed on a GCP VM with an Express.js backend and Next.js frontend.

JavaScript · Gemini · PostgreSQL · GCP · Next.js · Express.js

SuperGRT: GRAIN-QUALITY CV SYSTEM · LEAD · PATENT GRANTED

- Owned the full ML lifecycle (annotation, training, re-training and deployment) across grains, sugar, makhana, dry fruits and spices.
- Built classification for grain type, deep-learning segmentation for defect detection, and colorcard-based color grading for quality analysis.
- Replaced manual QC with a Jetson Nano running YOLOv8-Seg + TensorRT at 26 FPS, grading 200–300 grains/sample in ~15s vs. one or two by hand.
- Shipped APIs + backend for an Android inward-QC app (photo to cloud preprocessing to inference to grade in ~15s); standardized QC across 7 warehouses, trained on 15k custom images. Patent granted by the Indian Patent Office.

Python · PyTorch · ONNX · TensorRT · MLflow · NVIDIA DeepStream · YOLOv5/8 · Azure AI Studio

SuperHR: LLM HR AUTOMATION

- Cut HR ticket volume 70% with a chatbot + tool-calling agent for leave, payroll and policy over MySQL, with human approval loops on all database writes.
- Built a performance & goal-management module where managers and HR set, assign and track team goals, where an LLM drafts SMART goals, reviews and feedback summaries from KPIs, surfacing progress to managers.

JavaScript · OpenAI API · PostgreSQL · Fastify 5 · Next.js · Drizzle ORM

MLOps & Platform: INFRASTRUCTURE & ANALYTICS

- Cut model deployment time 60% and training cost 40% via Docker + MLflow + CI/CD on GCP.
- Built ML models for pricing, defect scoring and inventory prediction; shipped Power BI dashboards that cut ops reporting turnaround 80%.
- Built a D2C dashboard syncing orders, inventory and fulfillment across Flipkart, Amazon and CRED via API integration.

Data Science / Technology Intern · Khetika (SuperZop)

Jul 2022 – Jan 2023 · Mumbai

- Built a color + perspective correction module for warehouse phone images, improving downstream YOLO mAP ~5% and enabling the SuperGRT pilot.
- Wrote an image-QA service to auto-reject blurry/dark uploads, reducing bad data 90%.

Deep Learning Intern · Helppr.ai

Sep 2021 – Nov 2021 · Gurgaon

- Fine-tuned DistilBERT for 11-label UPSC subject classification: 95–98% accuracy, an 18% lift over baseline.

SKILLS

Languages: Python, SQL, JavaScript

LLM & Agents: LangChain, OpenAI API, Gemini API, RAG, NL-to-SQL, vector databases, tool-calling agents

Computer Vision: PyTorch, YOLOv8, Detectron2, ONNX, TensorRT, OpenCV, NVIDIA DeepStream

MLOps & Backend: Docker, MLflow, FastAPI, Flask, Streamlit, GitHub Actions, Kubernetes, Express.js, Fastify

Data & Cloud: PostgreSQL, MySQL, MongoDB, GCP, Azure VM, Azure Blob

ML & Analytics: Scikit-learn, Pandas, NumPy, regression, hypothesis testing, Power BI

ACHIEVEMENTS

- **Patent granted:** SuperGRT Grain Recognition Technology (Indian Patent Office).
- **FICCI Best Tech-Diffused Agri Startup Award (2025):** SuperTRACE.
- **Winner:** Internal AI Automation Hackathon (SuperTRACE); **Runner-up** (SuperHR).

EDUCATION

MSc, Computer Science (Data Science specialization): University of Mumbai

2019 – 2021

MSc, Physics: NES Ratnam College (University of Mumbai)

2018 – 2020

CERTIFICATIONS & LANGUAGES

Certifications: AWS Practical Data Science Specialization, Deep Learning Specialization, MLOps (ML Engineering for Production) Specialization, Machine Learning, Probability & Statistics

Languages: English, Hindi, Marathi